✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 1

Properties of Radio Waves

Radio Navigation & Aids — Ground Training Series

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1. Introduction

> **What this section covers:** The historical context of radio and radar in aviation, the transition from visual to instrument-based navigation, and why understanding radio fundamentals is essential for every pilot.

Radio and radar systems are now an integral and essential part of aviation, without which the current intensity of air transport operations would be unsustainable. In the early days of aviation, aircraft were flown with visual reference to the ground; flight at night, in cloud, or over the sea was not possible.

As aircraft complexity increased, it became necessary to design navigational systems that permit aircraft to operate without reference to terrain features. Early systems were very basic and inaccurate by modern standards — providing reasonable en route accuracy over land, but only limited service over the oceans. Until approximately 40 years ago, oceanic flight relied on traditional seafarer techniques of **astro-navigation** — taking sights on the sun, moon, stars, and planets to determine position.

Developments that commenced in the **1910s** accelerated during the **1930s and 1940s**, continuing to the present day, producing long-range systems that by the **1970s** provided a global navigation service.

> 🛩️ **Exam Perspective**
>
> It is perhaps ironic that, having forsaken navigation by the stars, the most widely used modern navigation systems — **Global Navigation Satellite Systems (GNSS)** — are once again space-based. The ICAO position (important for the exam) is that certain **terrestrial systems must be retained** to back up GNSS, both for en route navigation and runway approaches.

The development of radar in the **1930s** allowed air traffic control systems to be developed, providing a service capable of identifying and monitoring aircraft at a much higher intensity than would otherwise be possible. Modern satellite technology extends this capability over oceans and areas where conventional radar is not practical.

2. The Radio Navigation Syllabus

> **What this section covers:** An overview of the syllabus structure, explaining why the foundational chapters (radio wave properties, propagation, modulation, antennae) must be

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mastered before the applied navigation chapters.

The syllabus begins with the nature of radio waves and how they travel through the atmosphere. This is essential to understand:

- Why different radio frequencies are selected for particular applications
- The limitations those frequencies impose
- How radio waves are produced, transmitted, received, and how information is encoded and decoded

> ✅ **Syllabus Logic**
>
> Chapters 1–4 (waves → propagation → modulation → antennae) build the physics foundation. Chapters 5–18 apply that foundation to specific navigation and radar systems. The Q&A in Chapter 19 tests all chapters. Mastering Chapters 1–4 makes everything else easier to understand and retain.

3. Electromagnetic (EM) Radiation

> **What this section covers:** How radio waves are generated from an alternating current in a wire, the two-component nature of EM energy (electric and magnetic fields), and the concept of polarization.

If a **direct electric current (DC)** is passed through a wire, a magnetic field is generated around the wire perpendicular to the current flow.

If an **alternating electric current (AC)** is passed through the wire, because the direction of current flow is changing, the polarity of the magnetic field also changes — reversing polarity as the current direction reverses. At **low frequencies** the magnetic field collapses back to zero with the current; but as **frequency increases**, the magnetic field will not have collapsed completely before the reversed field starts to establish itself, and energy begins to travel outward from the wire in the form of **electromagnetic radiation** — i.e., radio waves.

The resulting EM energy is made up of two components:

- An **Electrical (E) field** — parallel to the wire
- A **Magnetic (H) field** — perpendicular to the wire

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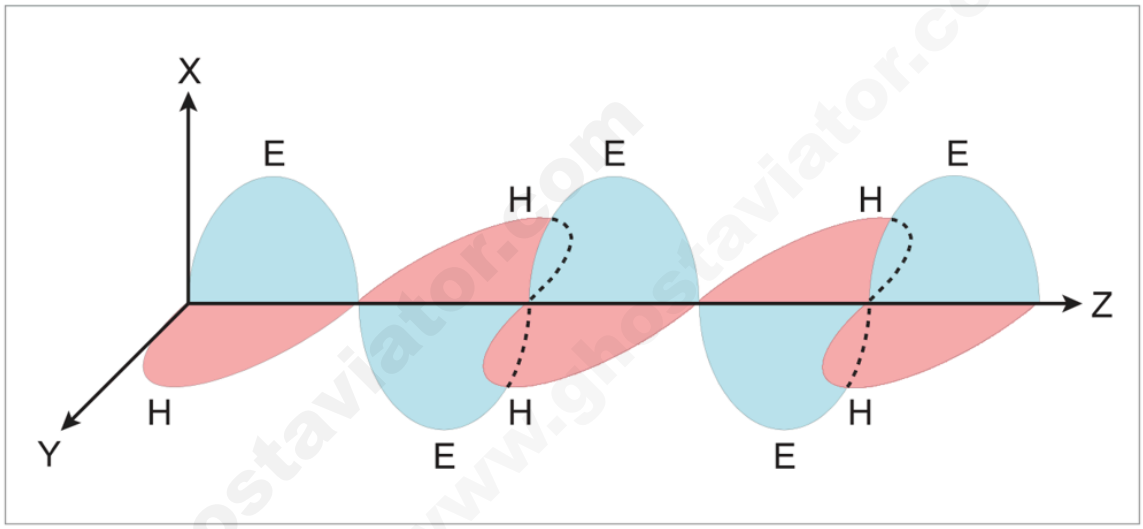


Figure 1.1 Vertical polarization

Figure 1.1 — Vertical polarization: E (electrical, blue) and H (magnetic, red) fields are perpendicular to each other, both propagating along the Z axis

Key Concept — The Two-Component Wave

A radio wave is NOT just an electrical or just a magnetic field — it is **both simultaneously, at right angles to each other and at right angles to the direction of propagation**. In Figure 1.1, blue lobes (E field) are in the vertical (X) plane; red lobes (H field) are in the horizontal (Y) plane; the wave travels along the Z axis. This perpendicular relationship is fundamental — it cannot be otherwise for EM radiation.

4. Polarization

What this section covers: Linear polarization (vertical and horizontal), circular polarization, the importance of aerial orientation for maximum signal strength, and the exam-relevant advantages of circular polarization in aviation.

The **polarization of radio waves** is defined as the plane of the *electric field* and is dependent on the plane of the aerial:

- A **vertical aerial** → electrical field in the vertical plane → **vertically polarized wave**
- A **horizontal aerial** → electrical field in the horizontal plane → **horizontally polarized wave**

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⚠️ **Critical Rule — Aerial Alignment**

To receive **maximum signal strength** from an incoming radio wave, the receiving aerial **must be in the same plane** as the polarization of the wave. A vertically polarized wave requires a vertical aerial. Misalignment between transmit and receive polarization causes significant signal loss.

Circular Polarization

Circular polarization can be produced in a variety of ways — one method uses a **helical antenna**. In circular polarization, the electrical (and hence magnetic) field **rotates at the frequency of the radio wave**. The rotation may be right-handed or left-handed, dependent on the orientation of the aerial array.

✅ **Two Key Advantages of Circular Polarization (Exam Favourite)**

1. **Rain clutter elimination in radar:** When circularly polarized energy is reflected from water droplets, the circularity of the returned signal is *reversed*. A receiver configured for the original rotation therefore rejects this return — eliminating precipitation clutter.
2. **Orientation-independent reception:** A simple dipole aerial can receive a circularly polarized wave regardless of its physical orientation. This is a **major advantage in mobile systems** — cellular phones, satellite communication, and satellite navigation (GNSS).

💡 **Exam Tip**

Questions on circular polarization almost always focus on one of these two advantages. Memorise: **"Rain clutter out, dipole orientation freed."**

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5. Radio Waves & The Period

What this section covers: The sinusoidal nature of radio waves, the period (τ), how frequency is derived from period, the units Hz/kHz/MHz/GHz, and how to perform frequency–wavelength conversions using the core formula.

The length of time it takes to generate **one complete cycle** of a radio wave is known as the **period**, signified by the Greek letter **tau (τ)**, and measured in **microseconds (μs)**.

$$1\ \mu s = 10^{-6}\ second$$

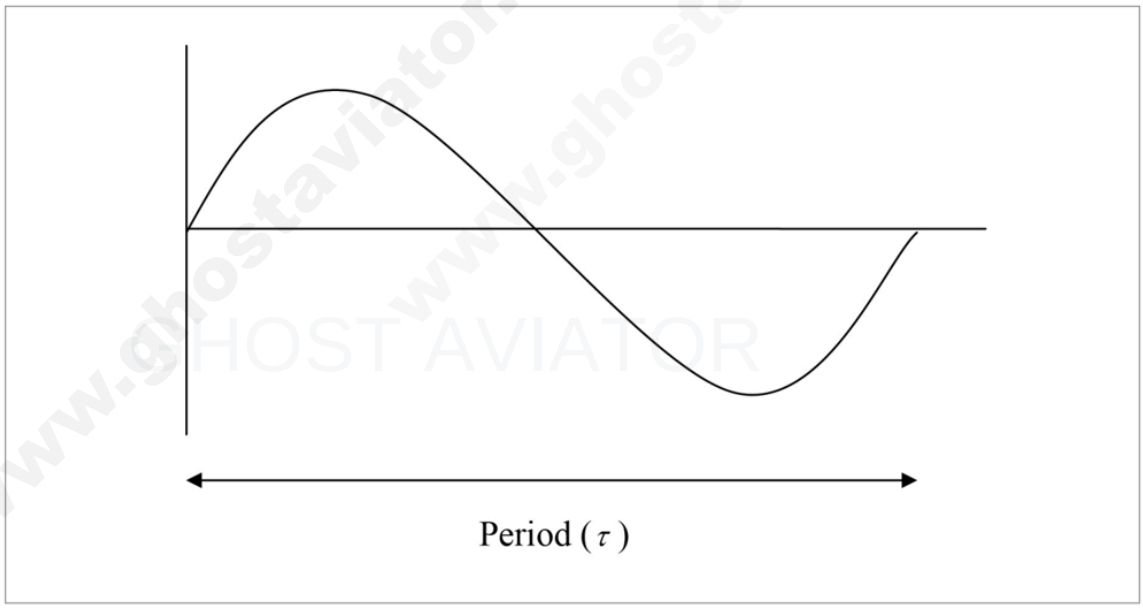


Figure 1.2 Sinusoidal wave - period

Figure 1.2 — Sinusoidal wave: the period (τ) is the time for one complete cycle, from zero through peak, trough, and back to zero

Frequency

Frequency is the reciprocal of period:

FORMULA 1 — FREQUENCY FROM PERIOD

$$f = 1 / \tau$$

where f = frequency (Hz) | τ = period (seconds)

📐 Worked Example — Period to Frequency

Given: $\tau = 0.125\ \mu s = 0.125 \times 10^{-6}\ s$

$f = 1 / (0.125 \times 10^{-6})$

$f = 8{,}000{,}000\ Hz = \mathbf{8\ MHz}$

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Frequency Units

Unit	Symbol	Value
Kilohertz	kHz	10^3 Hz = 1,000 Hz
Megahertz	MHz	10^6 Hz = 1,000,000 Hz
Gigahertz	GHz	10^9 Hz = 1,000,000,000 Hz

6. Wavelength

> **What this section covers:** The speed of radio waves, the wavelength formula, the simplified MHz–metre relationship, and worked examples for converting between frequency and wavelength.

The **speed of radio waves (c)** is the same as the speed of light (both are electromagnetic radiation):

$c =$ ***300,000,000 m/s*** $(= 300 \times 10^6\ m/s)$ *or* ***162,000 nautical miles per second***

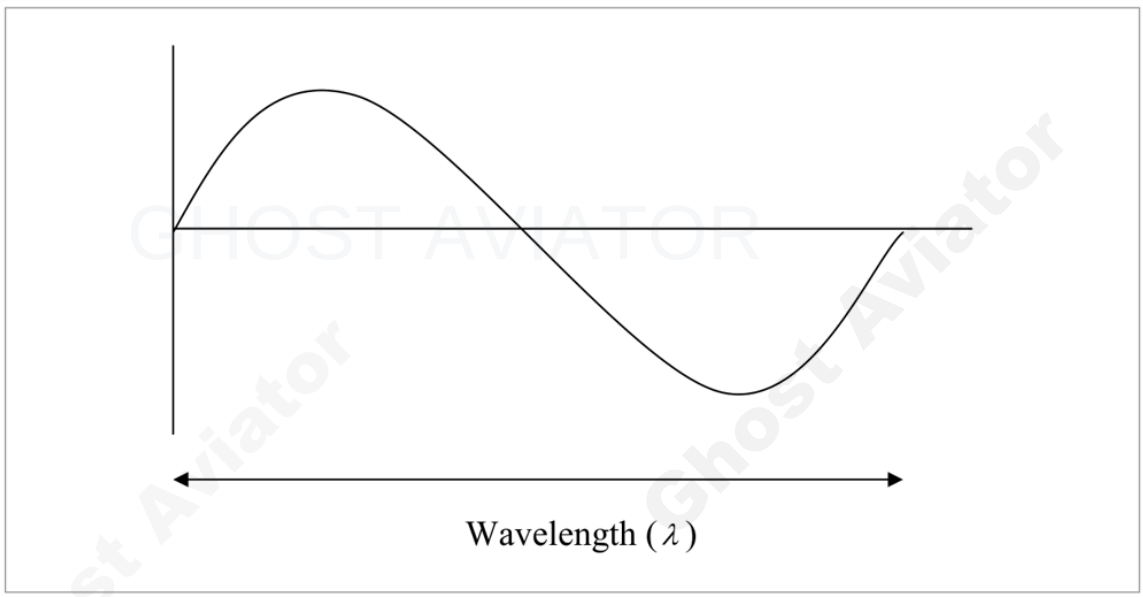


Figure 1.3 Sinusoidal wave - wavelength

Figure 1.3 — Wavelength (λ): the distance covered by one complete cycle of the wave at the speed of light

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FORMULA 2 — WAVELENGTH FROM PERIOD & SPEED

$$\lambda = c \times \tau \qquad (2)$$

$$\lambda = c / f \qquad (3)$$

$$f = c / \lambda \qquad (4)$$

λ = wavelength (m) | c = 300×10^6 m/s | f = frequency (Hz) | τ = period (s)

Simplified Formula (for Exam Speed)

SIMPLIFIED — MHZ & METRES

$$\lambda \text{ (m)} = 300 / f \text{ (MHz)}$$

$$f \text{ (MHz)} = 300 / \lambda \text{ (m)}$$

⚠ Input MUST be in MHz (for frequency) or metres (for wavelength). Convert units first!

📐 Worked Examples

Example 1: f = 121.5 MHz → λ = ?

λ = 300 / 121.5 = **2.47 m**

Example 2: λ = 1515 m → f = ?

f = 300 / 1515 = 0.198 MHz = **198 kHz**

Example 3: λ = 3.2 cm = 0.032 m → f = ?

f = 300 / 0.032 = 9375 MHz = **9.375 GHz**

Example 4: f = 357 kHz = 0.357 MHz → λ = ?

λ = 300 / 0.357 = **840 m**

⚠ Common Error — Unit Conversion

Always convert to **MHz and metres** before using the simplified formula. kHz must be divided by 1000. GHz must be multiplied by 1000. cm must be divided by 100. Skipping this step is the single most common calculation error in this chapter.

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7. Frequency Bands

> **What this section covers:** The 8 frequency bands of the radio spectrum (VLF to EHF), their frequency and wavelength ranges, and their civil aeronautical applications. This table is directly examinable.

The radio part of the electromagnetic spectrum extends from **3 kHz** to **300 GHz**. It is divided into 8 frequency bands, each related to its neighbours by a **factor of 10**.

Frequency Band	Abbreviation	Frequencies	Wavelengths	Civil Aeronautical Usage
Very Low Frequency	**VLF**	**3–30 kHz**	**100–10 km**	Nil
Low Frequency	**LF**	**30–300 kHz**	**10–1 km**	NDB / ADF
Medium Frequency	**MF**	**300–3000 kHz**	**1000–100 m**	NDB / ADF, long-range communications
High Frequency	**HF**	**3–30 MHz**	**100–10 m**	Long-range communications
Very High Frequency	**VHF**	**30–300 MHz**	**10–1 m**	Short-range comms, VDF, VOR, ILS Localizer, Marker Beacons
Ultra High Frequency	**UHF**	**300–3000 MHz**	**100–10 cm**	ILS Glidepath, DME, SSR, Satellite comms, GNSS, long-range radars
Super High Frequency	**SHF**	**3–30 GHz**	**10–1 cm**	RADALT, AWR (Weather Radar), MLS, short-range radars
Extremely High Frequency	**EHF**	**30–300 GHz**	**10–1 mm**	Nil

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💡 **Memory Aid — "Very Lovely Men Have Very Ugly Scruffy Enemies"**

VLF · **L**F · **M**F · **H**F · **V**HF · **U**HF · **S**HF · **E**HF

Key associations to memorise:

- NDB/ADF → **LF and MF**
- VOR, ILS Localizer, VHF Comms, Markers → **VHF**
- ILS Glidepath, DME, SSR, GNSS → **UHF**
- Weather Radar (AWR), RADALT, MLS → **SHF**



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8. Phase Comparison

What this section covers: The concept of phase, the requirements for a valid phase comparison (same frequency), how to determine phase difference graphically and mathematically, and why phase comparison is used in navigation systems such as VOR and ILS.

Some radio navigation systems use the **comparison of phase between two signals** to define navigational information. Two critical requirements:

⚠ **Two Absolute Requirements for Phase Comparison**

1. Both signals **must have the same frequency** — otherwise any phase comparison is meaningless.
2. The comparison must **yield a positive result** (0° to 360°). If a calculation gives a negative value, add 360°.

One signal is designated the **reference signal** and the other the **variable signal**.

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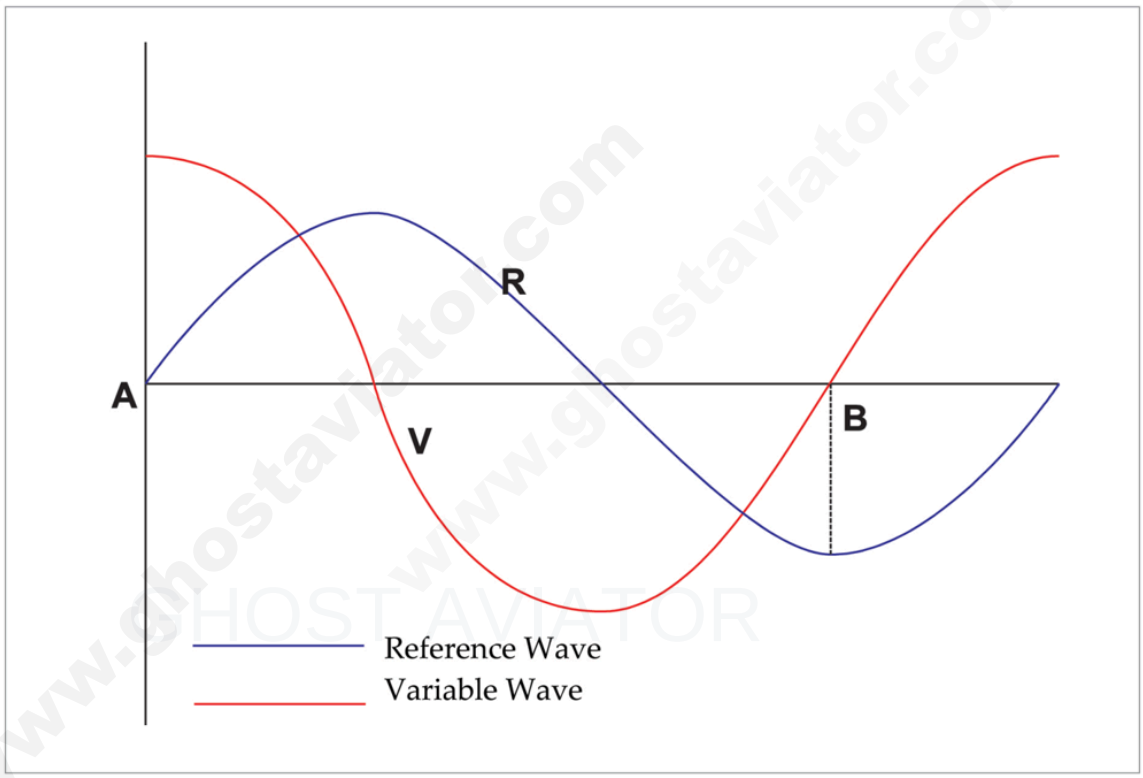


Figure 1.4 Sinusoidal wave - phase comparison

Figure 1.4 — Phase comparison: Reference Wave (blue/R) vs Variable Wave (red/V). Points A and B mark zero-phase crossing. Phase difference = 270°

Graphical Method

1. Identify zero-phase on the **reference wave** (point A in Figure 1.4)
2. Move in the **positive direction** from point A
3. Measure the phase angle the reference wave travels through before zero-phase is reached on the **variable wave** (point B)
4. That angle is the **phase difference**

In Figure 1.4: starting at zero phase on the reference wave (point A), the reference wave travels through **270°** before zero phase is reached on the variable wave (point B). Phase difference = **270°**.

Mathematical Method

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PHASE DIFFERENCE FORMULA

$$\text{Phase Difference} = \text{Phase}_{\text{Reference}} - \text{Phase}_{\text{Variable}}$$

Result must be positive. If negative, add 360°.

📐 Worked Example from Figure 1.4

At origin: Reference phase = 0° (= 360°)

Variable phase = 090°

Phase difference = 360° − 090° = **270°** ✓ (positive — valid)

📐 Worked Example — Negative Result Correction

Reference = 110° Variable = 315°

110° − 315° = −205° (negative — invalid)

Add 360°: −205° + 360° = **155°** ✓

💡 Exam Tip — Where is Phase Comparison Used?

Phase comparison is the fundamental operating principle of **VOR** (30 Hz reference vs 30 Hz variable), **ILS Localizer** (90 Hz vs 150 Hz amplitude), and **LORAN-C** (pulse phase comparison). Understanding this concept here makes those later chapters far easier.

9. Practice Frequency–Wavelength Conversions

What this section covers: Worked solutions to the 10 practice conversion problems from the source, with frequency band identification for each.

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#	Given	Find	Solution	Band
1	f = **198 kHz**	λ	300/0.198 = **1515 m**	LF
2	λ = **2.7 m**	f	300/2.7 = **111.1 MHz**	VHF
3	f = **5.025 GHz**	λ	300/5025 = **0.0597 m = 5.97 cm**	SHF
4	λ = **137.5 m**	f	300/137.5 = **2.182 MHz = 2181.8 kHz**	MF
5	f = **137.5 MHz**	λ	300/137.5 = **2.18 m**	VHF
6	λ = **3 km = 3000 m**	f	300/3000 = **0.1 MHz = 100 kHz**	LF
7	f = **329 MHz**	λ	300/329 = **0.912 m = 91.2 cm**	UHF
8	λ = **29 cm = 0.29 m**	f	300/0.29 = **1034 MHz**	UHF
9	f = **500 kHz**	λ	300/0.5 = **600 m**	MF
10	λ = **5 cm = 0.05 m**	f	300/0.05 = **6000 MHz = 6 GHz**	SHF

⚡ QUICK REVISION SUMMARY — CHAPTER 1

- EM wave = **E field (parallel to wire) + H field (perpendicular)**, both at 90° to propagation direction
- Polarization = plane of the **electric field**
- Maximum signal: receiving aerial must be in the **same plane as wave polarization**
- Circular polarization advantages: **rain clutter rejection** + **dipole orientation-free reception**
- Speed of radio waves: **300 × 10^6 m/s** or **162,000 NM/s**
- Key formulas: **f = 1/τ** | **λ = c/f** | simplified: **λ(m) = 300 / f(MHz)**
- Radio spectrum: **3 kHz to 300 GHz**, 8 bands
- Phase comparison requirement: **same frequency**, result must be **positive**
- ICAO position: **terrestrial systems must be retained** as GNSS backup

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📝 Practice Questions & Detailed Answers

How to Use This Section

Cover the answer panel and attempt each question independently. Reveal the answer, read the explanation, and follow the anchor link back to the relevant study section if you need to review the concept. The distractor analysis tells you exactly why each wrong option is wrong — that depth is where exam marks are won.

Q1. A radio wave is:

a. an energy wave comprising an electrical field in the same plane as a magnetic field

b. an electrical field alternating with a magnetic field

c. an energy wave where there is an electrical field perpendicular to a magnetic field

d. an energy field with an electrical component

✓ Correct Answer: **(c)** — an energy wave where there is an electrical field perpendicular to a magnetic field

Explanation: A radio wave is electromagnetic radiation comprising two components: an **E (electrical) field** and an **H (magnetic) field**. These two fields are always **perpendicular to each other**, and both are perpendicular to the direction of propagation. See Section 3 — EM Radiation.

Why the other options are wrong:

- **(a)** — The E and H fields are NOT in the same plane; they are perpendicular to each other. "Same plane" is the opposite of the correct relationship.

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- **(b)** — The E and H fields do not alternate with each other — they coexist simultaneously and continuously, both present at the same point in time, at 90° to each other.
- **(d)** — Incomplete and misleading. A radio wave has BOTH an electrical AND a magnetic component. Describing it as only having "an electrical component" misses the essential magnetic field.

Instructor's Note: *This question tests the very definition of EM radiation. Many students confuse "alternating with" (sequential) vs "perpendicular to" (simultaneous at 90°). Remember: BOTH fields exist at the same moment, at right angles — that is what makes EM radiation self-sustaining through space.*

Q2. The speed of radio waves is:

a. 300 km per second

b. 300 million metres per second

c. 162 NM per second

d. 162 million NM per second

✓ Correct Answer: **(b)** — 300 million metres per second

Explanation: The speed of radio waves equals the speed of light: **300,000,000 m/s = 300 × 10^6 m/s**. In nautical miles, this is **162,000 NM/s** — note 162 thousand, not 162 million. See Section 6 — Wavelength.

Why the other options are wrong:

- **(a)** — 300 km/s is vastly too slow. 300 km/s = 300,000 m/s. The correct value is 300,000,000 m/s — a factor of 1000 greater.
- **(c)** — 162 NM/s is off by a factor of 1,000. The correct NM value is 162,000 NM/s. This distractor is a classic exam trap.

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- **(d)** — 162 million NM/s would be faster than light, which is physically impossible. This is off by a factor of 1,000 in the wrong direction.

Instructor's Note: *Two correct values to memorise:* ***300,000,000 m/s*** *(or 300 × 10^6 m/s) AND* ***162,000 NM/s****. The NM figure appears directly in DME ranging calculations and radar questions later in the course.*

Q3. The plane of polarization of an electromagnetic wave is:

a. the plane of the magnetic field

b. the plane of the electrical field

c. the plane of the electrical or magnetic field dependent on the plane of the aerial

d. none of the above

✓ Correct Answer: **(b)** — the plane of the electrical field

Explanation: By definition, the polarization of a radio wave is the plane of the **electrical (E) field**. This is a fixed definition — it is always the E field, regardless of orientation. A vertical aerial produces a vertically polarized wave because the E field is vertical. See Section 4 — Polarization.

Why the other options are wrong:

- **(a)** — The magnetic field plane is always perpendicular to the E field. Polarization is defined by the E field, not the H field. This is a direct definition test.
- **(c)** — Plausible-sounding but wrong. The definition is NOT "dependent on the aerial plane" — it is always and definitively the E field. While the E field direction IS determined by the aerial orientation, the polarization is defined as the E field plane, full stop.
- **(d)** — Incorrect; option (b) is correct.

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Instructor's Note: *Pure recall: polarization = plane of the E field. Option (c) is the most common wrong answer because students confuse "the definition of polarization" with "how polarization is set". The aerial sets the E field plane; the E field plane is called the polarization.*

Q4. If the wavelength of a radio wave is 3.75 metres, the frequency is:

a. 80 kHz

b. 8 MHz

c. 80 MHz

d. 800 kHz

✓ Correct Answer: **(c)** — 80 MHz

Explanation: Using the simplified formula: f (MHz) = 300 / λ (m) = 300 / 3.75 = **80 MHz**. See Section 6 — Wavelength.

Why the other options are wrong:

- **(a)** — 80 kHz = 0.08 MHz. That would give a wavelength of 300/0.08 = 3750 m — three orders of magnitude too large.
- **(b)** — 8 MHz gives λ = 300/8 = 37.5 m, not 3.75 m. Off by a factor of 10.
- **(d)** — 800 kHz = 0.8 MHz → λ = 300/0.8 = 375 m. Completely different order of magnitude.

Instructor's Note: *3.75 m falls in the VHF band (1–10 m). Confirming the answer is in VHF (30–300 MHz) validates that 80 MHz is correct. Always sense-check your answer against the frequency band table.*

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Q5. The wavelength corresponding to a frequency of 125 MHz is:

a. 2.4 m

b. 24 m

c. 24 cm

d. 24 mm

✓ Correct Answer: **(a)** — 2.4 m

Explanation: λ = 300 / f (MHz) = 300 / 125 = **2.4 m**. 125 MHz is VHF, so wavelength should be in the 1–10 m range. 2.4 m fits perfectly. See Section 6.

Why the other options are wrong:

- **(b)** — 24 m would correspond to f = 300/24 = 12.5 MHz (HF band), not VHF.
- **(c)** — 24 cm = 0.24 m → f = 300/0.24 = 1250 MHz (UHF), not VHF.
- **(d)** — 24 mm = 0.024 m → f = 12.5 GHz (SHF). Nowhere near 125 MHz.

Instructor's Note: *The frequency 125 MHz is close to the VHF comms band (118–137 MHz). Knowing that VHF comms wavelengths are approximately 2–2.5 m is useful background knowledge that cross-validates this calculation.*

Q6. The frequency which corresponds to a wavelength of 6.98 cm is:

a. 4298 GHz

b. 4.298 GHz

c. 429.8 GHz

d. 42.98 GHz

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✓ Correct Answer: **(b)** — 4.298 GHz

Explanation: λ = 6.98 cm = **0.0698 m**. f = 300 / 0.0698 = 4297.9 MHz ≈ **4.298 GHz**. 6.98 cm is in the SHF band (1–10 cm), so frequency should be 3–30 GHz. 4.298 GHz fits. See Section 6 and Section 7.

Why the other options are wrong:

- **(a)** — 4298 GHz is in the EHF/infrared range. A factor of 1000 too large — forgetting to convert cm to m.
- **(c)** — 429.8 GHz is EHF range. A factor of 100 too large.
- **(d)** — 42.98 GHz is EHF range. A factor of 10 too large.

Instructor's Note: *The key trap:* ***must convert cm → m first****. 6.98 cm ÷ 100 = 0.0698 m. Only then apply f = 300/0.0698. Failing to convert is the #1 error on this type of question.*

Q7. The frequency band containing the frequency corresponding to 29.1 cm is:

a. HF

b. VHF

c. SHF

d. UHF

✓ Correct Answer: **(d)** — UHF

Explanation: λ = 29.1 cm = 0.291 m. f = 300/0.291 ≈ 1031 MHz ≈ **1.031 GHz**. UHF band spans 300–3000 MHz (0.3–3 GHz). 1031 MHz falls squarely in UHF. Alternatively: UHF wavelengths are 10–100 cm; 29.1 cm is in that range. See Section 7.

Why the other options are wrong:

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- **(a)** — HF = 3–30 MHz, wavelengths 10–100 m. 29.1 cm is far shorter than HF wavelengths.
- **(b)** — VHF = 30–300 MHz, wavelengths 1–10 m. 29.1 cm = 0.291 m is below the VHF wavelength range.
- **(c)** — SHF = 3–30 GHz, wavelengths 1–10 cm. 29.1 cm is too long for SHF.

Instructor's Note: *Alternative approach — use the wavelength ranges directly from the band table (no calculation needed): UHF = 10–100 cm. 29.1 cm is in 10–100 cm range → UHF. Knowing wavelength ranges as well as frequency ranges gives you two routes to the answer.*

Q8. To carry out a phase comparison between two electromagnetic waves:

a. both waves must have the same amplitude

b. both waves must have the same frequency

c. both waves must have the same amplitude and frequency

d. both waves must have the same phase

✓ Correct Answer: **(b)** — both waves must have the same frequency

Explanation: For a phase comparison to be meaningful, both signals **must have the same frequency**. If frequencies differ, the relative phase changes continuously — the comparison is meaningless. Amplitude is irrelevant to phase comparison. See Section 8 — Phase Comparison.

Why the other options are wrong:

- **(a)** — Amplitude has no bearing on phase comparison. In VOR, for example, the reference and variable signals can have different amplitudes but the same frequency — phase comparison still works.
- **(c)** — Same frequency is required, but same amplitude is NOT required. Adding amplitude as a requirement makes this option incorrect.

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- **(d)** — If both waves had the same phase, the phase difference would always be 0° — there would be nothing meaningful to compare. This option describes a trivial degenerate case, not a requirement.

Instructor's Note: *This is a direct recall question. The one and only requirement stated in the source: "the two signals being compared must have the same frequency, otherwise any phase comparison would be meaningless."*

Q9. The phase of the reference wave is 110° as the phase of the variable wave is 315°. What is the phase difference?

a. 205°

b. 025°

c. 155°

d. 335°

✓ Correct Answer: **(c)** — 155°

Explanation: Phase difference = Reference − Variable = 110° − 315° = −205°. Negative result → add 360°: −205° + 360° = **155°**. See Section 8.

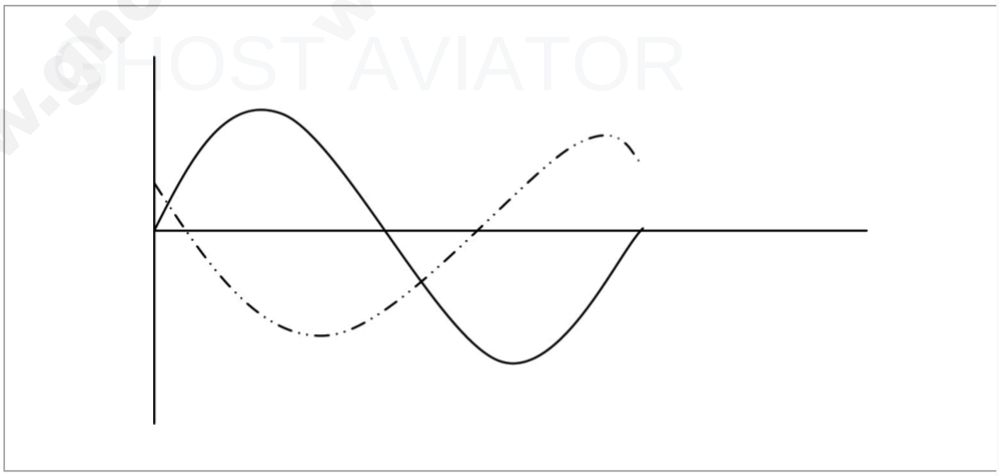
Why the other options are wrong:

- **(a)** — 205° is the absolute difference (315° − 110°) but calculated in the wrong direction. Phase difference is Reference minus Variable, not the reverse.
- **(b)** — 025° has no mathematical basis in this calculation.
- **(d)** — 335° = 315° + 20° — no valid derivation.

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> ***Instructor's Note:*** *Always do* ***Reference MINUS Variable****. If the result is negative, add 360°. Never do Variable minus Reference (that gives you 205° here — a common wrong answer). The formula direction matters.*

Q10. Determine the approximate phase difference between the reference wave and the variable wave: (The reference wave is the solid line and the variable wave is the dashed line)



Q10 Figure — Reference wave (solid) vs Variable wave (dashed). Determine phase difference.

a. 045°

b. 135°

c. 225°

d. 315°

✓ Correct Answer: **(c)** — 225°

Explanation: From the diagram, start at zero phase on the reference wave (solid line). Moving in the positive direction, the reference wave travels through approximately **225°** before zero phase is reached on the variable wave (dashed line). The variable wave appears to lead the reference by about 135°, which means

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the reference wave must travel 360° − 135° = 225° in the positive direction. See Section 8.

Why the other options are wrong:

- **(a)** — 045° is too small. The variable wave is not just slightly ahead of the reference.
- **(b)** — 135° would be the phase difference if you mistakenly measured Variable − Reference (in the negative direction). This reverses the formula.
- **(d)** — 315° would require the variable wave to be only 45° ahead of the reference — not what is shown in the diagram.

Instructor's Note: *On graphical phase questions, always identify the reference wave first, locate its zero crossing, then measure how far it travels (positive direction) before the variable wave's zero crossing. Don't measure the gap directly — measure the arc from the reference zero crossing in the forward direction.*

Q11. The wavelength corresponding to a frequency of 15,625 MHz is:

a. 1.92 m

b. 19.2 m

c. 1.92 cm

d. 19.2 cm

✓ Correct Answer: **(c)** — 1.92 cm

Explanation: λ = 300 / 15,625 = 0.0192 m = **1.92 cm**. 15,625 MHz = 15.625 GHz, which is in the SHF band (3–30 GHz). SHF wavelengths are 1–10 cm, so 1.92 cm is confirmed correct. See Section 6 and Section 7.

Why the other options are wrong:

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- **(a)** — 1.92 m = 0.00192 × 10^3 cm. This wavelength corresponds to 300/1.92 ≈ 156.25 MHz (VHF). Not SHF.
- **(b)** — 19.2 m would correspond to 300/19.2 ≈ 15.6 MHz (HF band). Completely wrong band.
- **(d)** — 19.2 cm = 0.192 m → f = 300/0.192 ≈ 1563 MHz (UHF). Wrong band — off by a factor of 10.

Instructor's Note: *15,625 MHz is a typical weather radar or airborne radar frequency (SHF band). Always check: SHF → wavelengths in centimetres (1–10 cm). This eliminates options in metres immediately.*

Q12. Which frequency band is a wavelength of 1200 m?

a. UHF

b. LF

c. HF

d. MF

✓ Correct Answer: **(b)** — LF

Explanation: f = 300 / 1200 m = 0.25 MHz = **250 kHz**. LF band spans **30–300 kHz**. 250 kHz falls in LF. Alternatively: LF wavelengths are 1–10 km; 1200 m = 1.2 km, which is in the 1–10 km range. Both routes confirm LF. See Section 7.

Why the other options are wrong:

- **(a)** — UHF wavelengths are 10–100 cm. 1200 m is completely different — six orders of magnitude longer than UHF wavelengths.
- **(c)** — HF band = 3–30 MHz, wavelengths 10–100 m. 1200 m is too long for HF (factor of ~12 too long).
- **(d)** — MF band = 300 kHz–3 MHz, wavelengths 100–1000 m. 1200 m is just above the upper MF wavelength limit of 1000 m, placing it in LF

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(which extends to 10 km). This is the most plausible wrong answer — 1200 m is close to the MF/LF boundary.

Instructor's Note: *The MF/LF boundary is at 300 kHz = 1000 m wavelength. 1200 m > 1000 m, so the frequency (250 kHz) is below 300 kHz → LF. Option (d) catches students who don't check the boundary carefully. Recall: longer wavelength = lower frequency = lower band.*

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📊 Master Reference Tables

All Numerical Values — Chapter 1

Value	Description	Section
3 kHz – 300 GHz	Full radio spectrum range	§7
300 × 10^6 m/s	Speed of radio waves (metres)	§6
162,000 NM/s	Speed of radio waves (nautical miles)	§6
3–30 kHz / 100–10 km	VLF band	§7
30–300 kHz / 10–1 km	LF band (NDB/ADF)	§7
300–3000 kHz / 1000–100 m	MF band (NDB/ADF, comms)	§7
3–30 MHz / 100–10 m	HF band (long-range comms)	§7
30–300 MHz / 10–1 m	VHF band (comms, VOR, ILS LOC)	§7
300–3000 MHz / 100–10 cm	UHF band (ILS GP, DME, SSR, GNSS)	§7
3–30 GHz / 10–1 cm	SHF band (AWR, MLS, RADALT)	§7
30–300 GHz / 10–1 mm	EHF band (nil aviation use)	§7

Formula Sheet

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Formula	Application	Units
f = 1/τ	Frequency from period	Hz, seconds
λ = c/f	Wavelength from frequency	m, m/s, Hz
f = c/λ	Frequency from wavelength	Hz, m/s, m
λ(m) = 300 / f(MHz)	Simplified (exam use)	metres, MHz
f(MHz) = 300 / λ(m)	Simplified (exam use)	MHz, metres
PD = Ref − Var (+ 360° if neg)	Phase difference	degrees

Answer Key — Chapter 1 Questions

Q1	Q2	Q3	Q4	Q5	Q6
c	b	b	c	a	b

Q7	Q8	Q9	Q10	Q11	Q12
d	b	c	c	c	b

Mnemonics & Memory Aids

- **Band order:** "Very Lovely Men Have Very Ugly Scruffy Enemies" → VLF · LF · MF · HF · VHF · UHF · SHF · EHF
- **Polarization:** "E defines the plane" — polarization is always the *Electric* field plane
- **Circular pol. advantages:** "Rain out, Dipole free" — rejects precipitation clutter, dipole receives regardless of orientation
- **Speed:** 300 million m/s = 162 thousand NM/s (never 162 million)
- **Phase difference:** Reference MINUS Variable → add 360° if negative
- **Simplified formula:** λ × f = 300 (when λ in metres, f in MHz)

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